

MAKUSHINO, Russian Federation

WMO: 286660

Lat: 55.250N

Lon: 67.300E

Elev: 459

StdP: 14.45

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.5	-21.4	-33.4	0.8	-27.3	-27.3	1.2	-21.2	25.8	21.9	22.4	13.0	4.0	200	0.643

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	88.3	66.7	85.2	66.3	81.9	65.1	69.7	82.4	68.3	80.7	67.0	78.6	8.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.2	74.0	64.1	91.2	72.8	62.6	86.2	72.3	34.0	82.0	32.8	81.0	31.8	78.6	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.4	21.0	18.3	-31.8	93.6	5.8	3.3	-35.9	95.9	-39.3	97.9	-42.5	99.7	-46.7	102.1
			-31.2	73.5	5.7	2.5	-35.3	75.3	-38.6	76.8	-41.8	78.2	-45.9	80.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	0.9	5.3	20.2	40.3	55.4	64.6	67.5	63.1	52.4	38.7	21.4	7.9
	DBStd	25.67	12.73	11.92	9.56	10.20	9.47	8.49	6.99	8.10	8.26	9.05	13.08	13.65
	HDD50	6674	1521	1251	924	321	54	6	0	3	69	362	858	1305
	HDD65	10685	1986	1671	1389	740	319	110	58	135	385	814	1308	1770
	CDD50	1804	0	0	0	31	223	443	543	410	141	13	0	0
	CDD65	339	0	0	0	0	22	97	136	77	7	0	0	0
	CDH74	3495	0	0	0	9	295	1025	1338	753	75	0	0	0
	CDH80	1103	0	0	0	1	79	346	423	243	11	0	0	0
Wind	WSAvg	8.5	8.2	8.6	9.2	9.6	9.0	8.3	7.0	7.4	8.3	8.8	8.9	9.0
Precipitation	PrecAvg	14.4	0.6	0.4	0.5	0.7	1.2	2.2	2.7	1.9	1.3	1.1	0.9	0.7
	PrecMax	21.7	1.3	1.1	1.8	1.9	3.0	4.7	6.3	3.9	4.1	3.9	1.7	2.6
	PrecMin	7.0	0.0	0.0	0.0	0.0	0.1	0.2	0.5	0.2	0.0	0.1	0.0	0.0
	PrecStd	3.2	0.3	0.3	0.4	0.5	0.7	1.3	1.5	0.9	0.9	0.9	0.4	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.0	34.8	46.8	74.0	87.2	91.4	92.2	90.7	81.8	66.8	46.9	35.3
		MCWB	30.7	31.8	40.9	55.7	63.2	67.0	68.3	68.1	59.6	52.4	42.4	32.5
	2%	DB	28.0	31.1	38.8	66.5	81.4	87.8	88.0	86.4	75.7	61.0	42.5	32.2
		MCWB	26.7	29.3	35.1	51.5	61.5	66.0	67.8	66.8	57.8	48.9	39.6	30.8
	5%	DB	23.9	27.4	36.0	61.4	76.8	84.3	84.5	82.4	70.5	56.2	39.3	29.2
		MCWB	22.8	25.8	33.2	49.6	59.6	66.2	67.2	65.6	57.0	47.0	36.7	27.7
	10%	DB	19.9	22.9	33.8	56.3	72.1	80.3	81.2	78.0	65.7	51.7	35.6	25.8
		MCWB	18.8	21.6	31.5	47.6	56.8	64.5	66.2	63.5	54.7	45.0	33.3	24.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.8	32.4	41.0	57.0	67.6	71.1	72.4	70.2	62.9	53.8	43.2	32.8
		MCDB	31.7	34.0	46.3	71.5	82.1	83.8	82.9	84.4	75.6	64.0	45.3	34.2
	2%	WB	26.8	29.3	35.9	53.2	63.2	68.8	70.4	68.4	59.9	50.9	40.1	30.8
		MCDB	28.0	31.1	38.2	62.1	77.7	82.3	81.8	81.8	70.9	58.8	42.4	32.4
	5%	WB	22.9	25.9	33.5	50.7	60.6	66.9	68.9	67.0	57.7	48.0	36.8	27.9
		MCDB	23.9	27.3	35.6	61.0	73.6	80.2	80.6	78.5	67.8	54.7	39.1	29.2
	10%	WB	18.9	21.6	31.6	47.7	58.2	65.3	67.6	65.2	55.7	45.5	33.7	24.4
		MCDB	19.9	23.0	33.7	56.1	70.1	78.1	78.8	75.5	64.7	51.3	35.1	25.8
Mean Daily Temperature Range	5% DB	MDBR	13.0	15.0	15.2	15.9	20.3	19.5	19.4	18.8	16.9	13.4	10.9	12.6
		MCDBR	13.6	15.0	13.6	23.9	26.3	25.1	24.2	24.4	23.9	21.2	10.9	12.4
		MCWBR	13.1	14.0	10.8	12.5	11.6	10.0	8.8	9.2	11.1	12.2	9.2	11.7
	5% WB	MCDBR	13.5	15.1	12.7	22.3	23.3	22.7	21.0	22.2	21.3	19.1	10.4	12.4
		MCWBR	13.1	14.2	10.6	12.5	11.9	10.3	9.2	9.4	11.0	11.9	9.2	11.9
Clear-Sky Solar Irradiance	taub	0.282	0.317	0.370	0.411	0.407	0.404	0.424	0.433	0.383	0.357	0.318	0.284	
	taud	2.054	2.020	1.957	2.112	2.231	2.313	2.255	2.195	2.292	2.232	2.121	2.019	
	Ebn at Noon	193	230	244	255	265	267	258	246	244	216	177	158	
	Edh at Noon	21	32	43	43	41	38	40	40	31	26	20	17	
All-Sky Solar Radiation	RadAvg	297	633	1130	1445	1754	1891	1776	1418	967	488	239	187	
	RadStd	21	30	72	153	140	173	148	142	76	80	31	26	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page